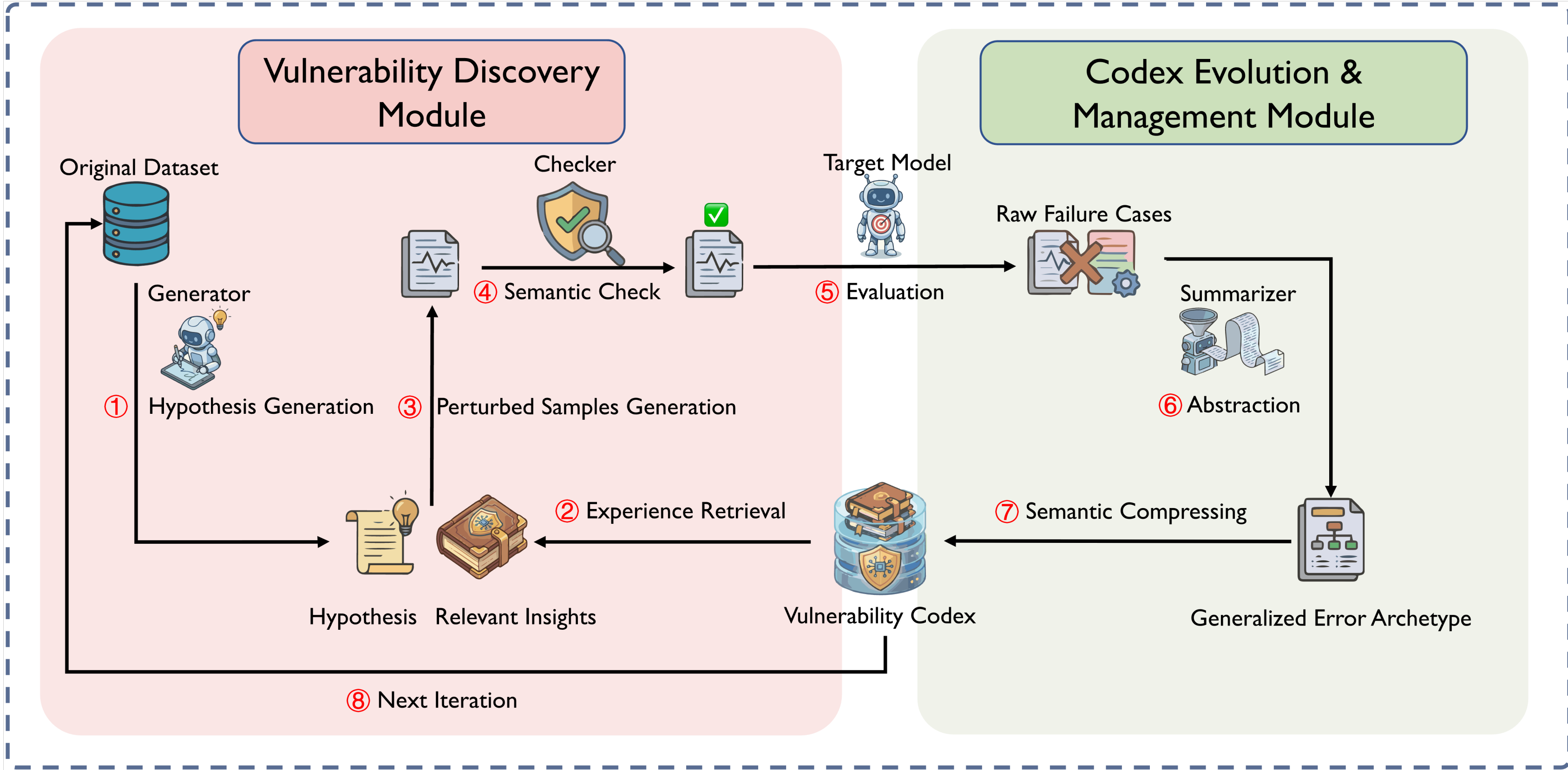


Background & motivation

- Real-world Text-to-SQL reliability remains underexamined.
- Static benchmarks use fixed test sets.
- Rule-based perturbations rely on human-defined error types.
- SAGE: Systematic Automated Guided Exploration to uncover latent failure patterns on LLM-based Text-to-SQL generations.

SAGE: Systematic Automated Guided Exploration



Perturbation example

Q: Name 10 users with the badge name 'Necromancer.'
Evidence: users refers to DisplayName (unchanged)
Unused field (1 of 5): postHistory.UserDisplayName
Before: "... associated with the post editor."
After: "... may be system-generated ... or manually set."

Failure case

SELECT UserId FROM badges
WHERE Name = 'Necromancer' **LIMIT** 10;
Before: 10 name rows [whuber, whuber, conjugateprior, ...]
After: 10 ID rows [919, 919, 1739, ...]

Codex example

errorType: QUESTION_INTERPRETATION_ERROR
errorSummary: "Misunderstood the intent to retrieve a human-readable user identifier (DisplayName) rather than a foreign key (UserId)"

I Main Results

- Gemma-3, Inf-rl-qwen and OmniSQL on BIRD and Spider; static Expert baseline.
- SAGE surpasses Expert: higher VER and lower ApD.

Model	Setting	BIRD				Spider			
		EX	VES	VER	ApD	EX	VES	VER	ApD
Gemma-3	Original	53.65	63.75	-	-	81.53	109.35	-	-
	Expert	24.64	29.38	54.07	7.73	48.36	61.83	52.16	10.95
	SAGE	8.34	9.86	84.45	5.08	12.38	15.72	66.32	7.85
Inf-rl-qwen	Original	70.53	87.03	-	-	87.72	116.38	-	-
	Expert	38.20	44.87	45.84	9.81	63.93	82.11	27.12	19.97
	SAGE	21.90	26.03	68.95	7.27	56.19	71.05	48.07	12.93
OmniSQL	Original	63.23	74.77	-	-	81.83	110.85	-	-
	Expert	30.31	36.63	52.06	7.09	48.84	63.57	40.10	10.60
	SAGE	14.21	17.06	77.53	5.47	31.82	40.35	60.97	8.15

EX: execution accuracy (%).VES: valid efficiency score.VER: vulnerability exposure rate (%).ApD: attempts per discovery. Source: Table 1.

2 Cross-Model Discovered Vulnerability Generalization Analysis

- Cells: VER gain over Expert (pp); source rows, target columns.
- All cross-model gains are positive: transferable structural weaknesses.

Codex / target	Gemma-3	Inf-rl-qwen	OmniSQL
Gemma-3	+18.47	+25.23	+13.61
Inf-rl-qwen	+18.23	+10.91	+12.27
OmniSQL	+16.89	+25.69	+15.16

3 Mitigating Vulnerability with Fine-tuning

- SAGE generates 1,580 adversarial samples.
- Use these samples to fine-tune Qwen2.5-Coder-7B-Instruct.
- Adversarial EX +5.21 pp; standard BIRD EX +10.24 pp.

Training	BIRD dev EX	Adversarial EX	ApD
Base	42.63	4.37	3.61
+ SAGE SFT	52.87 (+10.24)	9.58 (+5.21)	4.59

Novel patterns

- Schema Misinterpretation
- Syntactic Overfitting
-

Conclusion

- SAGE's self-evolving framework outperforms static expert rules.
- Vulnerabilities discovered by SAGE are transferable across models.
- SAGE can generate targeted training data.
- The idea behind SAGE can be applied to other tasks and domains.